

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 October 2021 Worked Solutions

October 2021 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 October 2021

October 2021 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Binomial Expansion

1. The first three terms, in ascending powers of x , of the binomial expansion of $(1 + kx)^{16}$ are

$$1, -4x \text{ and } px^2$$

where k and p are constants.

- (a) Find, in simplest form,

(i) the value of k

(ii) the value of p

(3)

$$g(x) = \left(2 + \frac{16}{x}\right)(1 + kx)^{16}$$

Using the value of k found in part (a),

- (b) find the term in x^2 in the expansion of $g(x)$.

(3)

(Total 6 marks)

Leave
blank

Worked Solution - Question 1

Topic group

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: - = -

Final answer

- = -

Question 2

Sequences & Series

$$u_1 = 6$$
$$u_{n+1} = ku_n + 3$$

where k is a positive constant.

- (a) Find, in terms of k , an expression for u_3

(2)

Given that $\sum_{n=1}^3 u_n = 117$

- (b) find the value of k .

(3)

(Total 5 marks)

Worked Solution - Question 2

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: therefore = 7 2 k =

Final answer

therefore = 7 2 k =

Question 3

Integration

3.

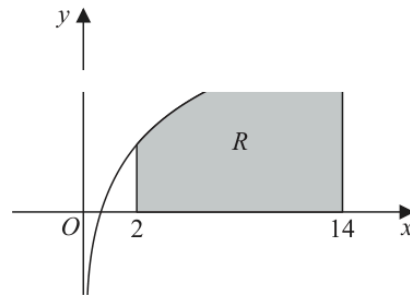


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = \log_{10} x$

The region R , shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 2$, the x -axis and the line with equation $x = 14$

Using the trapezium rule with four strips of equal width,

(a) show that the area of R is approximately 10.10 (3)

(b) Explain how the trapezium rule could be used to obtain a more accurate estimate for the area of R . (1)

(c) Using the answer to part (a) and making your method clear, estimate the value of

(i) $\int_2^{14} \log_{10} \sqrt{x} \, dx$

(ii) $\int_2^{14} \log_{10} 100x^3 \, dx$

(4)

(Total 8 marks)

Worked Solution - Question 3

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: $+ \times = \int$

Final answer

$$+ \times = \int$$

Question 4

Polynomials

(a) State the value of the remainder when $f(x)$ is divided by $(2x - 3)$

(1)

(b) Use the factor theorem to show that $(x - 3)$ is a factor of $f(x)$

(2)

(c) Hence,

(i) factorise $f(x)$

(ii) show that the equation $f(x) = 0$ has only one real root.

(5)

(Total 8 marks)

Worked Solution - Question 4

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: $3x = 0$

Final answer

$$3x = 0$$

Question 5

Modelling with Sequences & Series

- extracted 480 tonnes of silver from the mine in year 1
- extracted 465 tonnes of silver from the mine in year 2
- extracted 450 tonnes of silver from the mine in year 3

and so on, forming an arithmetic sequence.

- (a) Find the mass of silver extracted in year 14

(2)

After a total of 7770 tonnes of silver was extracted, the company stopped mining.

Given that this occurred at the end of year N ,

- (b) show that

$$N^2 - 65N + 1036 = 0$$

(3)

- (c) Hence, state the value of N .

(1)

(Total 6 marks)

Worked Solution - Question 5

Topic group

1. Model the situation

Match the real situation to an arithmetic, geometric, or recurrence model using the starting value and rate of change.

2. Interpret the result

Calculate the required term or sum and state the answer in the units and context of the question.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 6

Circles

$$x^2 + y^2 + 10x - 12y = k \quad \text{where } k \text{ is a constant}$$

- (a) Find the coordinates of the centre of C_1 (2)
- (b) State the possible range in values for k . (2)
- (ii) The point $P(p, 0)$, the point $Q(-2, 10)$ and the point $R(8, -14)$ lie on a different circle, C_2

Given that

- p is a positive constant
- QR is a diameter of C_2

find the exact value of p . (4)

(Total 8 marks)

Worked Solution - Question 6

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: 3 165 p = + only

Final answer

3 165 p = + only

Question 7

Geometric Sequences

Leave
blank

7. (i) A geometric sequence has first term 4 and common ratio 6
Given that the n^{th} term is greater than 10^{100} , find the minimum possible value of n .
(3)

- (ii) A different geometric sequence has first term a and common ratio r .

Given that

- the second term of the sequence is -6
- the sum to infinity of the series is 25

- (a) show that

$$25r^2 - 25r - 6 = 0$$

(3)

- (b) Write down the solutions of

$$25r^2 - 25r - 6 = 0$$

(1)

Hence,

- (c) state the value of r , giving a reason for your answer,
(1)

- (d) find the sum of the first 4 terms of the series.
(2)

(Total 10 marks)

Worked Solution - Question 7

1. Use the geometric formula

Identify the first term, common ratio, and whether the question needs a term, finite sum, or sum to infinity.

2. Substitute and solve

Apply the correct geometric formula and simplify, checking any validity condition on the ratio.

3. State the final result

Write the answer in the form requested by the question: = 24

Final answer

= 24

Question 8

Integration

Solutions relying entirely on calculator technology are not acceptable.

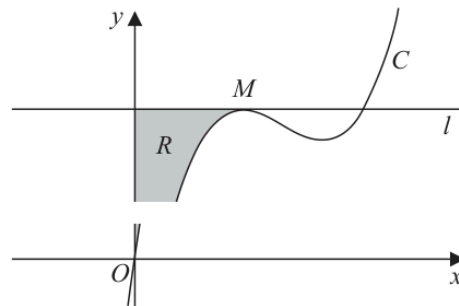


Figure 2

Figure 2 shows a sketch of part of the curve C with equation

$$y = \frac{4}{3}x^3 - 11x^2 + kx \quad \text{where } k \text{ is a constant}$$

The point M is the maximum turning point of C and is shown in Figure 2.

Given that the x coordinate of M is 2

(a) show that $k = 28$ (3)

(b) Determine the range of values of x for which y is increasing. (2)

The line l passes through M and is parallel to the x -axis.

The region R , shown shaded in Figure 2, is bounded by the curve C , the line l and the y -axis.

(c) Find, by algebraic integration, the exact area of R . (5)

(Total 10 marks)

Worked Solution - Question 8

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: $x - x - x + x \ 40 \ 3 =$

Final answer

$$x - x - x + x \ 40 \ 3 =$$

Question 9

Proof

$$\frac{x+y}{2} \geq \sqrt{xy}$$

(3)

- (b) Prove by counter-example that this inequality does not hold when x and y are both negative.

(1)

(Total 4 marks)

Worked Solution - Question 9

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: $x + y$

Final answer

 $x + y$

Question 10

Trigonometric Equations

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

$$\tan^2\left(2x + \frac{\pi}{4}\right) = 3$$

(5)

- (ii) Solve, for $0 < \theta < 360^\circ$

$$(2 \sin \theta - \cos \theta)^2 = 1$$

giving your answers, as appropriate, to one decimal place.

(5)

Q10

(Total 10 marks)

(Total 10 marks)

Worked Solution - Question 10

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: degrees degrees degrees θ

Final answer

degrees degrees degrees θ